



Koneru Lakshmaiah Education Foundation

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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE TITLE & CODE : DATA SCIENCE (24AD3107) A.Y. 2026 - 2027

Title of the project	Sales & Demand Forecasting System Using ARIMA and Prophet	
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Abstract

The Sales & Demand Forecasting System Using ARIMA and Prophet is a Data Science project designed to predict future sales and product demand using historical sales data. Accurate sales forecasting is important for businesses to improve inventory management, production planning, and decision-making while reducing problems such as overstocking, understocking, and loss of sales opportunities. The system accepts sales data from CSV or Excel files and performs automated data validation and preprocessing before generating forecasts.

The proposed system includes an automated data cleaning pipeline to handle missing values, invalid records, and incorrect data types. It also performs data quality scoring based on completeness, uniqueness, validity, and consistency. Exact and similarity-based duplicate detection are used to identify repeated or near-duplicate records, while anomaly detection techniques such as IQR and Z-score identify unusual sales values. The processed data is further analyzed to identify sales trends, seasonal patterns, product performance, and demand behavior.

For forecasting, the system uses ARIMA (AutoRegressive Integrated Moving Average) and Prophet time-series models. Both models are trained using historical sales data and evaluated using MAE, RMSE, and MAPE. The system compares their performance and automatically selects the model with better forecasting accuracy. The final results are presented through an interactive dashboard containing data quality reports, sales analytics, historical versus forecast visualizations, forecast summaries, predicted demand, and business insights. This system

provides an end-to-end solution that transforms raw sales data into meaningful forecasts to support better business planning and decision-making.

Technology Stack

1. Programming Language

- **Python** – Main language for data processing, forecasting models, and application development.

2. Data Processing

- **Pandas** – Data cleaning, transformation, and analysis.
- **NumPy** – Numerical calculations and data manipulation.
- **OpenPyXL** – Reading Excel files.

3. Machine Learning & Forecasting

- **Statsmodels** – Implementing the **ARIMA** forecasting model.
- **Prophet** – Time-series forecasting with trend and seasonality.

4. Data Visualization

- **Plotly** – Interactive charts and graphs.
- **Matplotlib** – Additional data visualizations.

5. Dashboard / Frontend

- **Streamlit** – Interactive web dashboard for uploading data, viewing analytics, and forecasts.

6. Data Input

- **CSV**
- **Excel (.xlsx)**

Justification: We selected this project because sales and demand forecasting is a real-world business problem. Incorrect demand estimation can cause overstocking or understocking. Our system uses historical data, improves its quality through automated preprocessing, and applies

ARIMA and Prophet models to predict future demand. By comparing both models and selecting the more accurate one, the system provides reliable forecasts and helps businesses make better inventory and planning decisions.

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Project Guide